

Taller 3

⑤

a) Muestre que las matrices de Pauli, forman base para ese espacio vectorial

$$H_2 = \{ A \in M_2(\mathbb{C}) : A^* = A \}$$

• Para que las matrices de Pauli sean base, deben generar a todo H_2 y ser L.I

1) Veamos si $\text{gen}(\sigma_0, \sigma_1, \sigma_2, \sigma_3) = H_2 \rightarrow$ Sean $\sigma_0, \sigma_1, \sigma_2, \sigma_3 \in H_2$ y $\alpha_1, \alpha_2, \alpha_3, \alpha_4 \in \mathbb{R}$ note que

$$\alpha_1 \sigma_0 + \alpha_2 \sigma_1 + \alpha_3 \sigma_2 + \alpha_4 \sigma_3 = M_2(\mathbb{C})$$

$$\alpha_1 \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} + \alpha_2 \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} + \alpha_3 \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} + \alpha_4 \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} = \begin{pmatrix} z_1 & z_2 \\ z_3 & z_4 \end{pmatrix}$$

$$\begin{pmatrix} \alpha_1 + \alpha_4 & \alpha_2 - \alpha_3 i \\ \alpha_2 + \alpha_3 i & \alpha_1 - \alpha_4 \end{pmatrix} = \begin{pmatrix} z_1 & z_2 \\ z_3 & z_4 \end{pmatrix}$$

Ahora note

$$\begin{pmatrix} z_1^* & z_3^* \\ z_2^* & z_4^* \end{pmatrix} = \begin{pmatrix} \alpha_1 + \alpha_4 & \alpha_2 - \alpha_3 i \\ \alpha_2 + \alpha_3 i & \alpha_1 - \alpha_4 \end{pmatrix} = \begin{pmatrix} z_1 & z_2 \\ z_3 & z_4 \end{pmatrix} \in H_2$$

$$\Rightarrow \text{gen}(\sigma_1, \sigma_2, \sigma_3, \sigma_4) = H_2$$

• Veamos si $\sigma_0, \sigma_1, \sigma_2, \sigma_3$ son L.T $\rightarrow$ Sean $\alpha_1, \alpha_2, \alpha_3, \alpha_4 \in \mathbb{R}$

$$\alpha_1 \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} + \alpha_2 \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} + \alpha_3 \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} + \alpha_4 \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}$$

$$\begin{aligned} \alpha_1 + \alpha_4 &= 0 \\ \alpha_2 - \alpha_3 i &= 0 \\ \alpha_1 - \alpha_4 &= 0 \\ \alpha_2 + \alpha_3 i &= 0 \end{aligned} \Rightarrow \alpha_1 = 0, \alpha_2 = 0, \alpha_3 = 0, \alpha_4 = 0$$

b) Compruebe que esa base es ortogonal bajo la definición de producto interno

Para que sean ortogonales debe cumplir

$$\langle a | b \rangle = \text{Tr}(A^\dagger B)$$

$$\langle \sigma_i | \sigma_j \rangle = \delta_{ij} \|\sigma_i\|^2 //$$

$$\langle \sigma_i | \sigma_j \rangle = \text{Tr}(\sigma_i^\dagger \sigma_j) = \text{Tr}(\sigma_i \sigma_j)$$

Note que:

$$\text{Tr}(\sigma_i) = 0 \quad // \text{ como } \sigma_0 = I_2$$

$$\text{Tr}(\sigma_0 \sigma_i) = \text{Tr}(\sigma_i \sigma_0) = \text{Tr}(\sigma_i) = 0 = \langle \sigma_0 | \sigma_i \rangle = \langle \sigma_i | \sigma_0 \rangle$$

$$\langle \sigma_0 | \sigma_0 \rangle = \text{Tr}(\sigma_0 \sigma_0) = \text{Tr}(I) = 2$$

$$\langle \sigma_1 | \sigma_1 \rangle = \text{Tr}(\sigma_1 \sigma_1) = \text{Tr} \left[\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \right] = \text{Tr} \left[\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \right] = 2$$

$$\langle \sigma_1 | \sigma_2 \rangle = \text{Tr}(\sigma_1 \sigma_2) = \text{Tr} \left[\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} \right] = \text{Tr} \left[\begin{pmatrix} i & 0 \\ 0 & -i \end{pmatrix} \right] = 0$$

$$\langle \sigma_2 | \sigma_1 \rangle = \text{Tr}(\sigma_2 \sigma_1) = \text{Tr} \left[\begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \right] = \text{Tr} \left[\begin{pmatrix} -i & 0 \\ 0 & i \end{pmatrix} \right] = 0$$

$$\langle \sigma_1 | \sigma_3 \rangle = \text{Tr} \left[\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \right] = \text{Tr} \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} = 0 \quad \langle \sigma_3 | \sigma_2 \rangle = 0$$

$$\langle \sigma_3 | \sigma_3 \rangle = 2$$

$$\langle \sigma_3 | \sigma_1 \rangle = \text{Tr} \left[\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \right] = \text{Tr} \left[\begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} \right] = 0$$

$$\langle \sigma_2 | \sigma_2 \rangle = \text{Tr} \left[\begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} \right] = \text{Tr} \left[\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \right] = 2$$

$$\langle \sigma_2 | \sigma_3 \rangle = \text{Tr} \left[\begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \right] = \text{Tr} \left[\begin{pmatrix} 0 & i \\ i & 0 \end{pmatrix} \right] = 0$$

$$\Rightarrow \sigma_i \sigma_j = 2\delta_{ij} \Rightarrow \text{la base es ortogonal} //$$

c) Explora si puede construir subespacios vectoriales de matrices reales e imaginarias puras

Sea $|A\rangle \in H_2$ noto que

$$|A\rangle = \alpha_1 |\sigma_0\rangle + \alpha_2 |\sigma_1\rangle + \alpha_3 |\sigma_2\rangle + \alpha_4 |\sigma_3\rangle$$

$$|A\rangle = \underbrace{\alpha_1 |\sigma_0\rangle + \alpha_2 |\sigma_1\rangle + \alpha_4 |\sigma_3\rangle}_{\text{Subespacio 1 } S_1} + \underbrace{\alpha_3 |\sigma_2\rangle}_{\text{Subespacio 2 } S_2}$$

Sea $|A_1\rangle \in S_1 : S_1 \subseteq H_2$ y $\alpha_1, \alpha_2, \alpha_3 \in \mathbb{R}$

$$|A_1\rangle = \alpha_1 |\sigma_0\rangle + \alpha_2 |\sigma_1\rangle + \alpha_3 |\sigma_3\rangle$$

$$|A_1\rangle = \alpha_1 \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} + \alpha_2 \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} + \alpha_3 \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

$$|A_1\rangle = \begin{pmatrix} \alpha_1 + \alpha_3 & \alpha_2 \\ \alpha_2 & \alpha_1 - \alpha_3 \end{pmatrix} \Rightarrow \text{En } S_1 \subseteq H_2 \text{ est\u00e1n las matrices con entradas reales}$$

Sea $|A_2\rangle \in S_2 : S_2 \subseteq H_2$ y $\alpha \in \mathbb{R}$

$$|A_2\rangle = \alpha |\sigma_2\rangle = \alpha \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} = \begin{pmatrix} 0 & -\alpha i \\ \alpha i & 0 \end{pmatrix} \Rightarrow \text{En } S_2 \subseteq H_2 \text{ est\u00e1n las matrices con entradas imaginarias.}$$

$$\Rightarrow H_2 = S_1 \oplus S_2 //$$